

Introduction to Algorithms

L02 - Flowcharts

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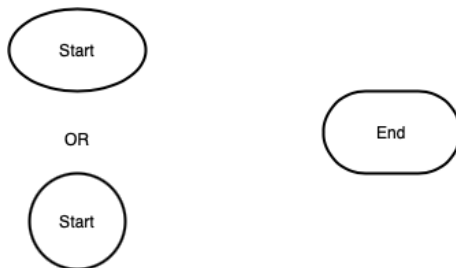
- With algorithmic thinking, we decompose a problem into parts.
- These parts may be defined using patterns that we already know.
- Designing the interaction between the parts/patterns results in an abstraction.
- A visual representation of the abstraction makes it easier to translate and develop the algorithm in a programming language.

- Flowcharts provide a visual abstraction of how the parts/patterns interact.
- This visual representation enables both programmers and non-programmers to understand the logic behind the algorithm.
- The act of breaking down a problem into an algorithm promotes problem solving.

ISO 5807 provides a list of common flowchart symbols. These include:

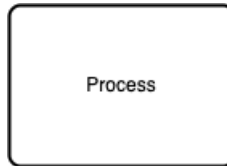
- Start/End*
- Processing*
- Predefined process*&**
- Decision*
- Input/Output*
- Flow arrows*
- Annotations or (notes)
- Connectors

Start/End symbols



Represents where to begin reading the flowchart and where the algorithm being processed will end. Some flowchart models do not include the begin and/or end symbols.

Processing symbol



Identifies where operations and data manipulations are applied.

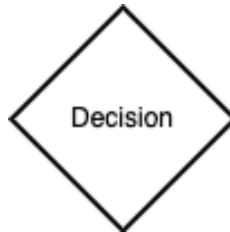
Predefined process symbol



Predefined Process

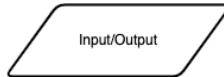
An encapsulation of a number of steps that are not specified for the particular behaviour being modelled by the rest of the given flowchart. For example, the steps to a pattern are not provided, but may be expanded on later.

Decision symbol



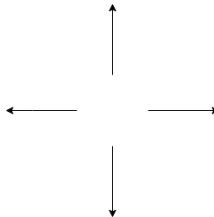
Represents an operation where there are 2 or more alternatives. The resulting flows are labelled to indicate the alternative options.

Input/Output symbol



Used to model input from the user or another source as well as to where the output will be piped.

Flow arrows

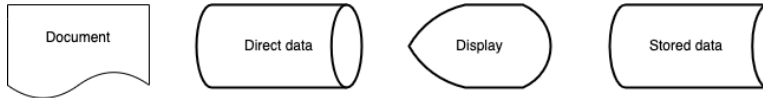


Connects symbols and indicates the direction at which information flows.

Other, lesser used symbols have been defined in the ISO standard. Examples of these include:

- Document
- Display
- Stored Data
- Database
- ...

Other Symbols

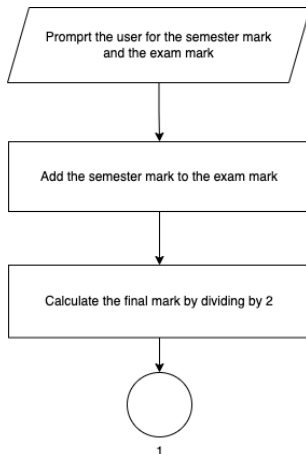


There are many other symbols. In most cases tools and users modelling algorithms using flowcharts use the basic symbols.

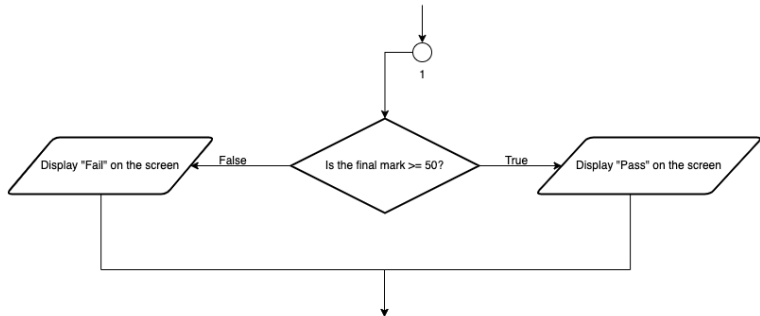
Flowchart structures are high-level patterns to illustrate:

- linear execution, i.e. sequence structures
- decision making, i.e. selection structures
- repeating a number of times until a condition is met, i.e. repetition structures

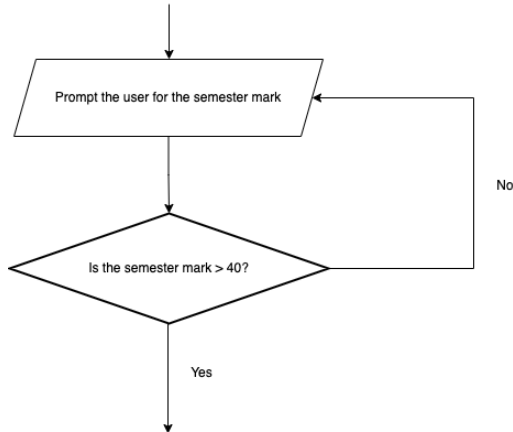
Sequence structure



Selection structure



Repetition structure



- Flowcharts must always be readable
- Do not let the flowchart to spread over many pages, keep is concise
- Keep fonts, arrows, text sizes etc. consistent.

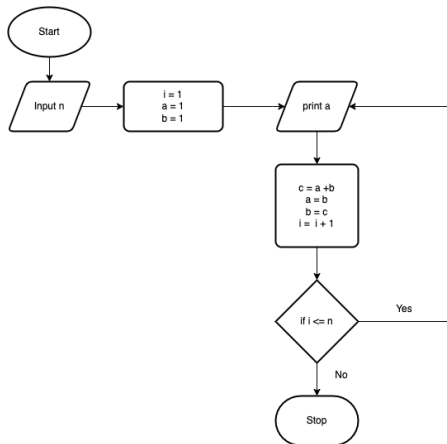
Students should be able to:

- Read and analyse existing flowcharts
- Be able to apply the steps provided in a flowchart when debugging code

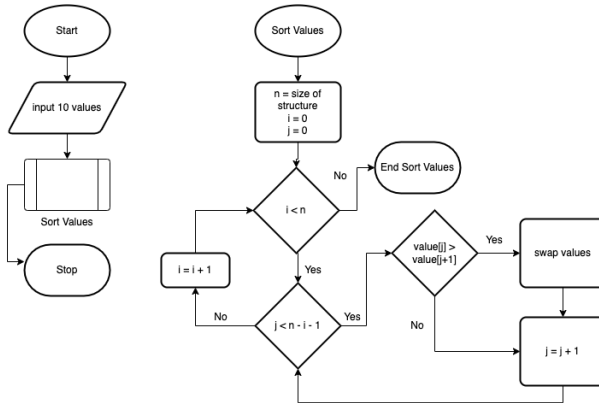
- Look for tools that help you to draw flowcharts.
- Some tools are static, while others simulate the execution of an algorithm represented by the flowchart.

- Figures in this presentation were drawn using *draw.io*. There is an online version as well as a desktop version
- *PseudoFlow* is online tool for pseudo code to flowchart conversion.
- Examples of build animation/executable flowchart tools are *AlgoBuild Home* and *RAPTOR*.

The following flowchart is a representation for which algorithm?



Sorting as a predefined process.



Consider the two examples from L01:

- Boiling the kettle; and
- Escaping the canyon.

Draw flowcharts representing the abstraction of the algorithms for each of these examples.